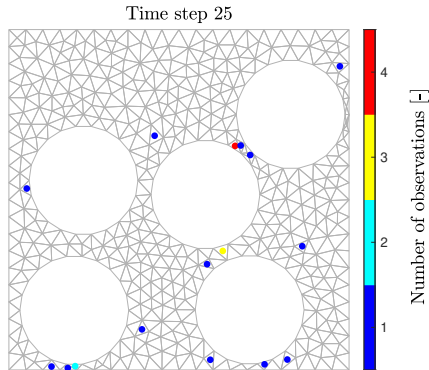
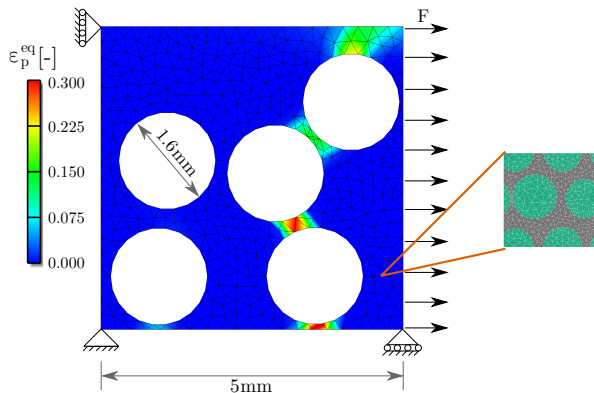


Active learning with GPs

Adaptive construction of the constitutive database:

- The model naturally concentrates sampling effort where it is needed



Towards hybrid physics-ML models

Pure machine learning has its drawbacks:

- MLPs: Curse of dimensionality, excessive variance
- GPs: Awkward for path dependency

Why does hyper-reduction extrapolate so well?

- Some material points are still computed with **purely physics-based models**
- Galerkin-based approach enforces PDE physics in reduced solution space

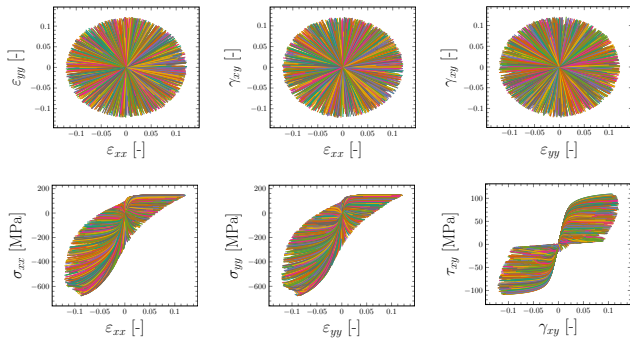
Can we transfer some of these insights back into machine learning?

- **Hybrid models!**

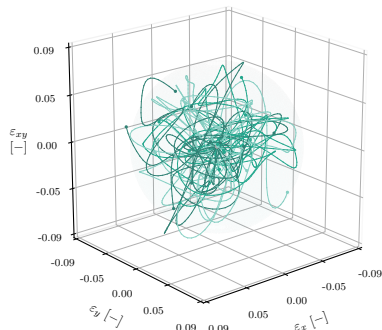
Sampling, training and testing

Monotonic/proportional strain paths or GP-based paths:

- Predicting unloading without training
- Comparing learning curves against modern RNNs

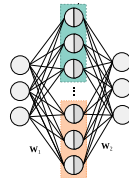


[Rocha et al. (2023), *Mech Mater*, In press]



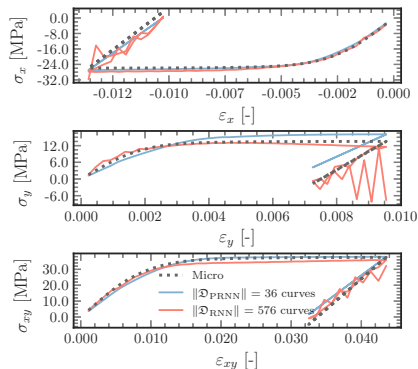
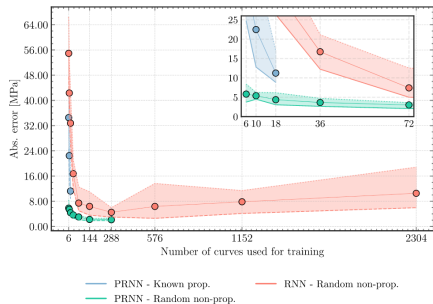
[Maia et al. (2023), *CMAME*, 407:115934]

RNNs versus PRNNs



Training both networks on GP paths:

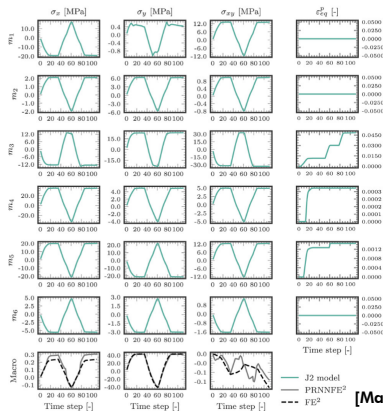
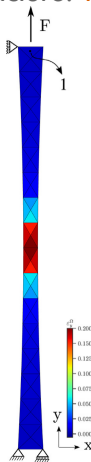
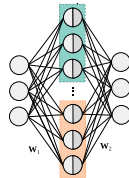
- Efficient learning through well-informed bias



PRNNs for FE²

Fast and interpretable, speedup of about **30000 times**

- Extension to fracture: **tomorrow** by Nóra Kovács (DDF-III, 14:40)



[Maia et al. (2023), CMAME, 407:115934]